

From Dialogue Systems to Chatbots

Sentiment and Emotional Framing in Conversational AI Research

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Research Question

How did conversational-AI terminology and emotion-associated language change from 1976 to 2025?

Background

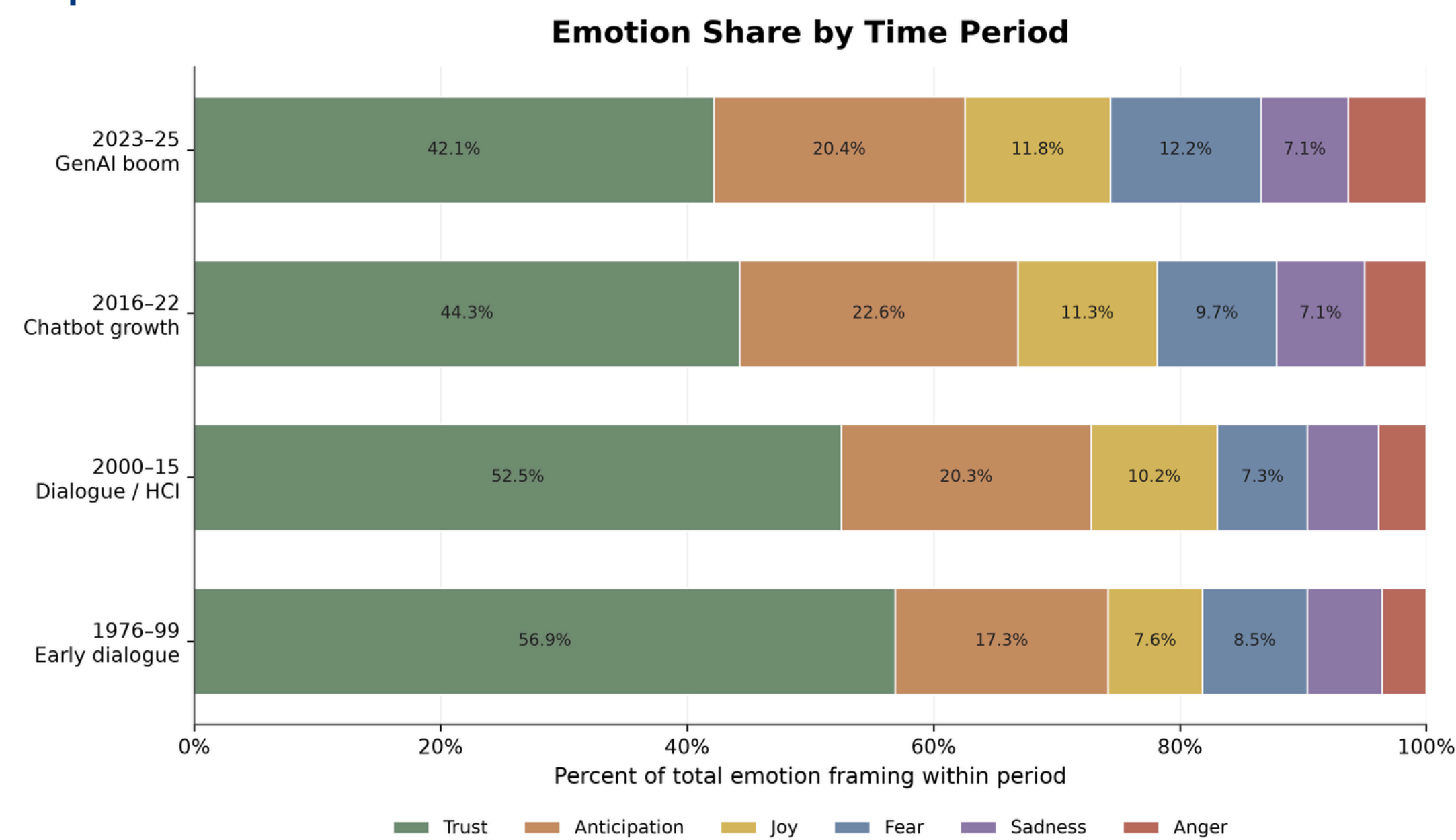
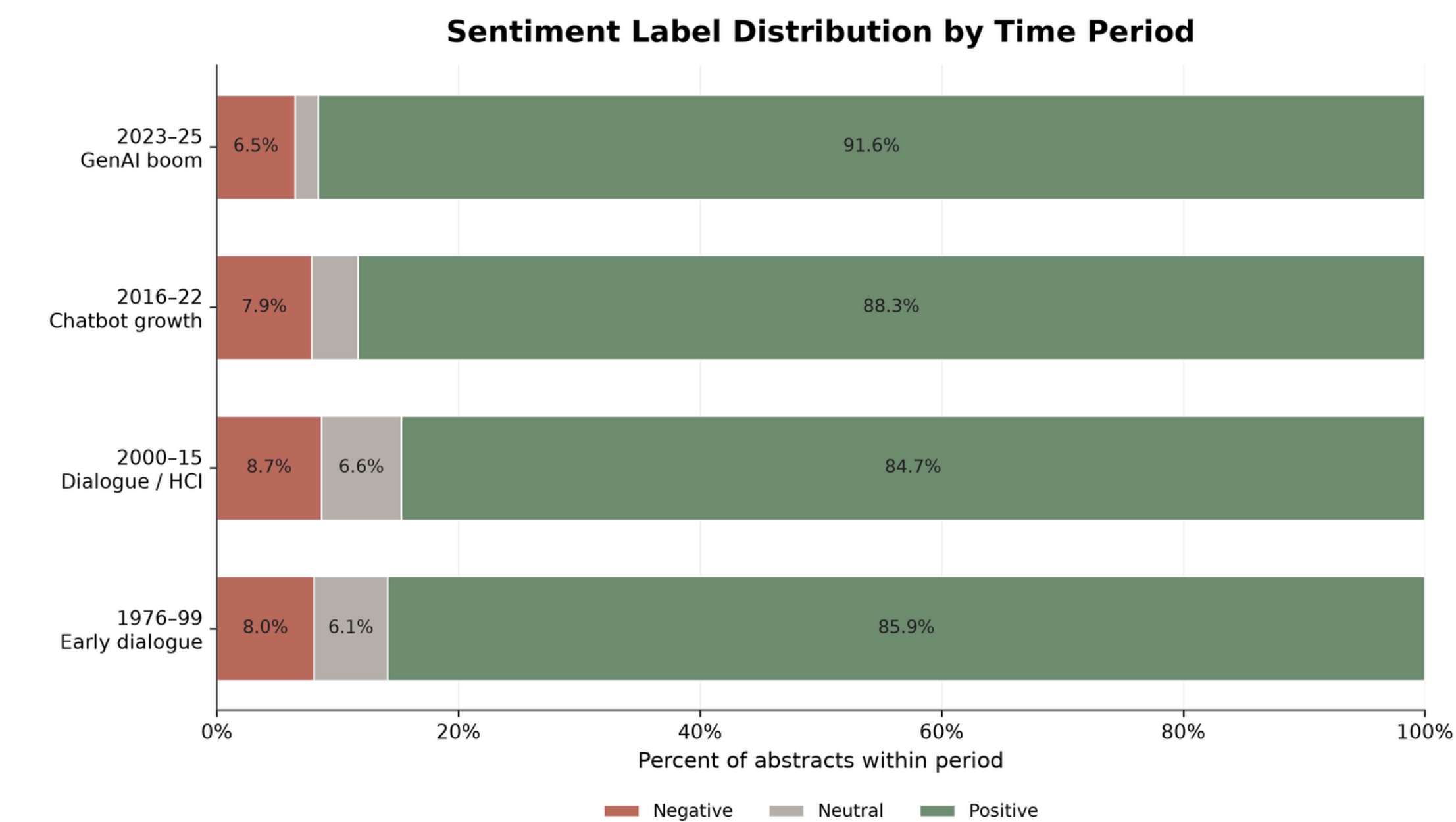
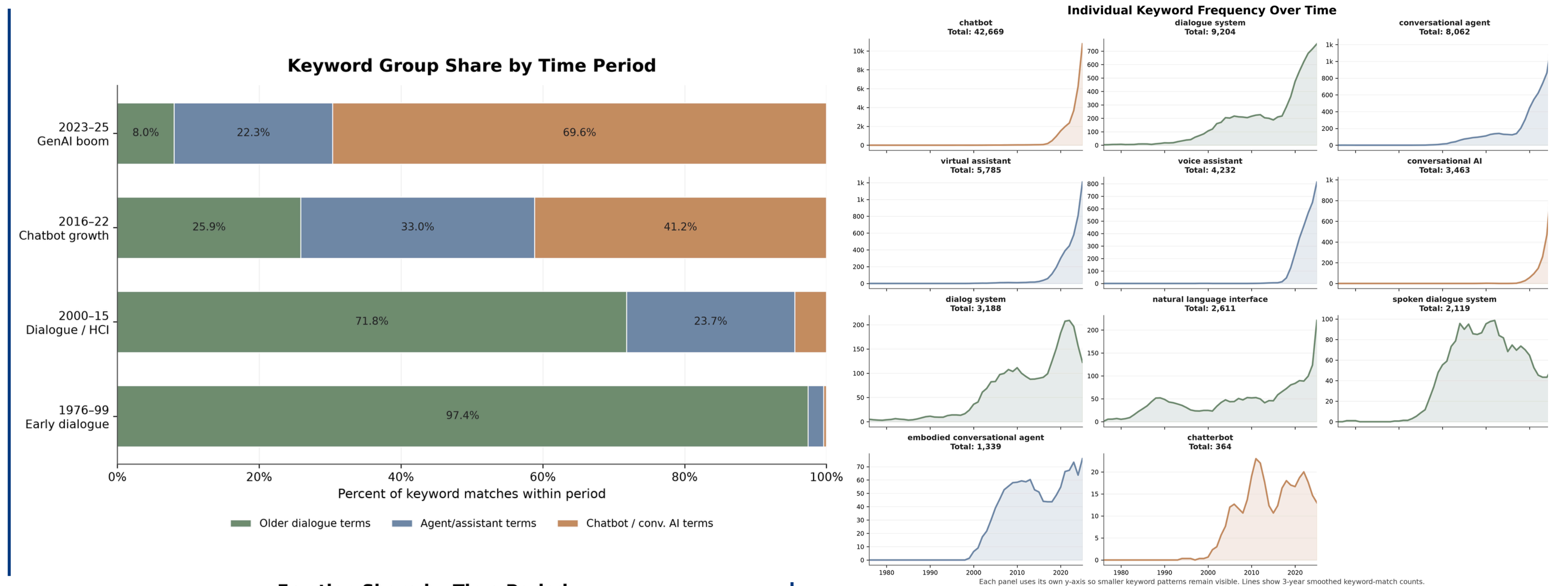
Earlier research described conversational systems using terms such as dialogue system, spoken dialogue system, and natural language interface. More recent studies increasingly use conversational agent, virtual assistant, chatbot, and conversational AI. This vocabulary shift must be considered when comparing sentiment and emotion-associated language over time.

Methods

-OpenAlex pipeline: Queried 11 search phrases, retained title or abstract matches, and cleaned and deduplicated the results. The final corpus contained 72,843 records, including 52,448 English abstracts used for text analysis.

-Lexicon scoring: VADER generated compound sentiment scores, while NRClex measured associations with six selected emotion categories.

-Grouped analysis: Three keyword groups were compared across four historical periods. Statistics represent keyword-record matches, so one paper could contribute to multiple groups.



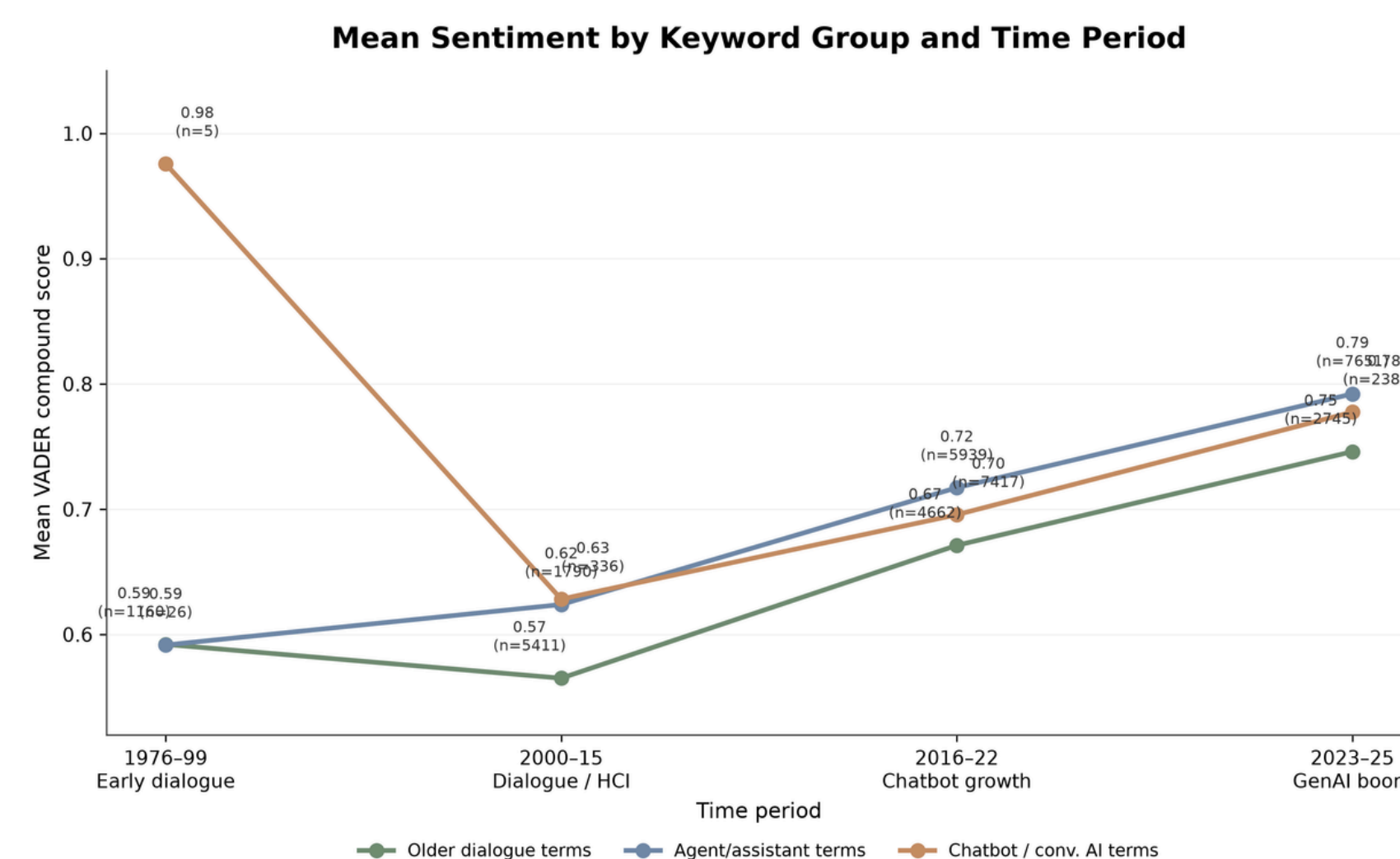
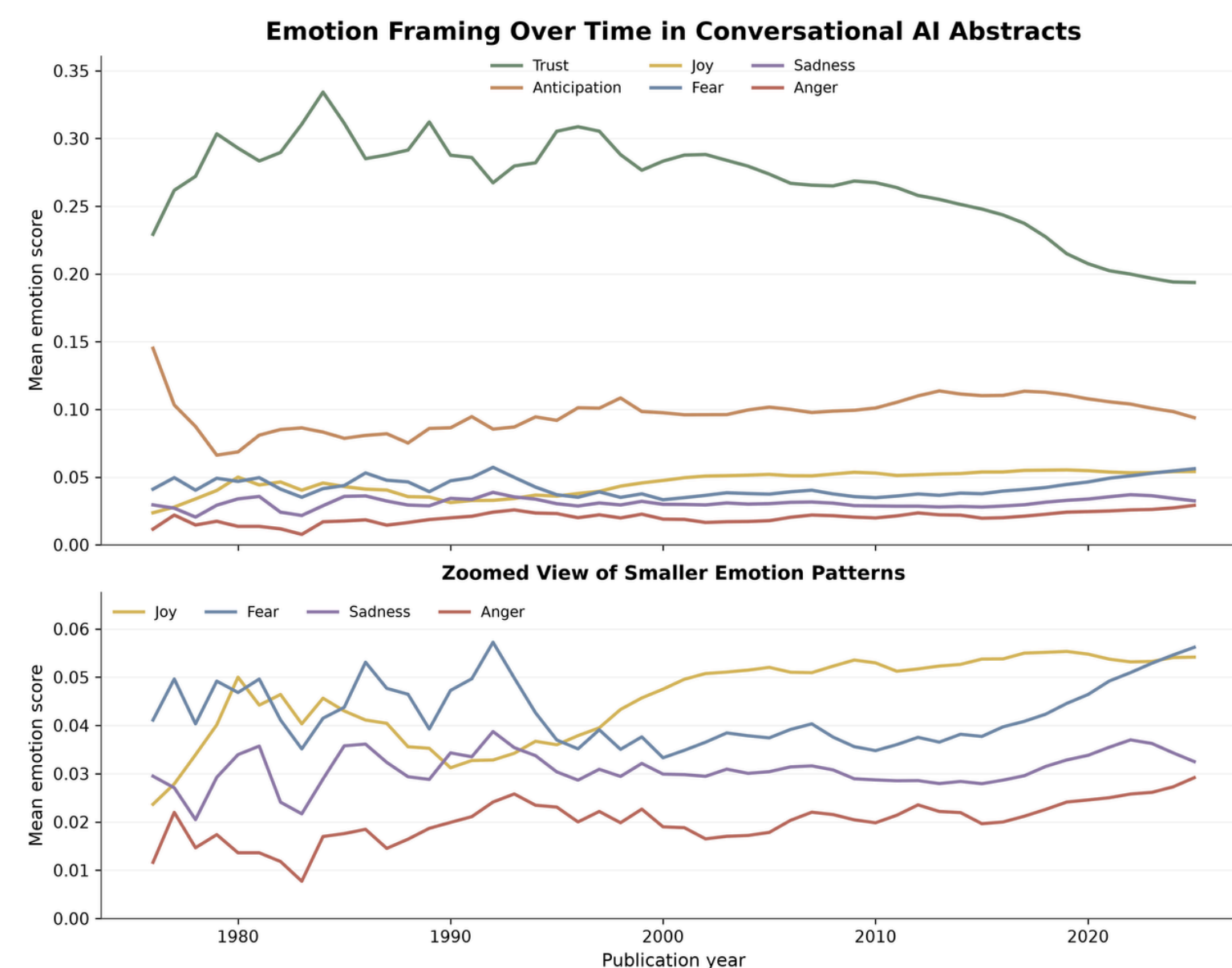
Results

Terminology: Older dialogue terminology declined from 97.4% of grouped keyword matches in 1976-1999 to 8.0% in 2023-2025, while chatbot/conversational-AI terminology increased to 69.6%. Across the full corpus, chatbot produced 42,669 keyword matches.

Sentiment: VADER classified 47,098 abstracts (89.8%) as positive, compared with 3,746 (7.1%) negative and 1,604 (3.1%) neutral. Period-level positivity ranged from 84.7% to 91.6%.

Emotion Associations: Among six selected NRC categories, trust's normalized share declined from 56.9% to 42.1%, while fear's share increased from 8.5% to 12.2% between the earliest and most recent periods.

Sentiment by Keyword Group: From 2016-2025, mean VADER compound scores ranged from 0.67 to 0.79 across keyword groups. The early chatbot/conversational-AI score of 0.98 was based on only five keyword-match rows and should be interpreted cautiously.



Conclusion

The corpus shows a clear shift from older dialogue-system terminology (97.4% of matches in 1976-1999) to chatbot/conversational-AI terminology (69.6% in 2023-2025), while 89.8% of abstracts received positive VADER labels. Among the six selected NRC categories, trust's normalized share declined from 56.9% to 42.1% as fear increased from 8.5% to 12.2%.

These are lexical associations, not measurements of author emotion.

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**LURF Sentiment
Analysis: Project
Repository**

